A million cars, two gigabytes of RAM, and five questions

Every job on VEIQ's fleet runs under a hard memory cap. Go over it and your job dies, not the machine. So the constraint below isn't us being clever, it's just our Tuesday. Five real market questions, one budget.

THE DATA

`case-c_market.parquet`: **1,000,000 Dutch listings** across ten marketplaces, first registrations 2005 to 2026, live and expired. 18 columns, 19 MB on disk, and about **750 MB** if you load it into pandas without thinking.

`listing_id` · **`seller_ref`** · **`seller_type`** `(dealer / private)` · `source` · `brand` · `model` · `first_registration_year` · **`asking_price_eur`** · `mileage_km` · `body_type` · `fuel_type` · `transmission` · `power_kw` · `province` · **`first_seen`** · **`last_seen`** · **`still_active`** · `status_raw`

The file is **synthetic**. We generated it to match the shape of the Dutch market rather than scraping it, so the cars aren't real. Every pathology in there is.

> **The constraint: 2 GB of RAM, and the whole thing runs in under 5 minutes on your laptop.**
> Measure both and tell us the numbers. Any language, any library: pandas, Polars, DuckDB, plain SQL, your own streaming loop. Loading all million rows at once is allowed if you can stay inside the budget. Usually you can't.

THE FIVE QUESTIONS

01 **Supply.**
How many distinct cars are actually on the market? The same car gets listed on several marketplaces at once, so you'll need to decide what counts as a duplicate, defend that call, and give us the deduplicated number.

02 **Liquidity.**
Which segments sell fastest? Use `first_seen`, `last_seen` and `still_active`. Watch out: a listing that's live today has no end date yet, so averaging the finished ones will lie to you.

03 **Price by age.**
Build a depreciation curve per fuel type: price against age, controlled for mileage. Tell us what "controlled for" means in whatever method you pick.

04 **Who is cheap.**
Do private sellers undercut dealers on comparable cars? Put a number on the gap and say what you counted as comparable.

05 **Something is wrong.**
This data is messy on purpose. Find at least three things wrong with it, show us the evidence, and say which of your four answers above they mess up.

THE TRAP WE'RE WATCHING FOR

Question 5 is the one that separates people. You can compute all four answers above perfectly and still be wrong, because the data underneath them is broken in places. Duplicate listings inflate supply. Relisted cars fake quick sales. €1 "call me" ads wreck any average. We'd rather see **four answers you trust and one you've flagged as unreliable** than five confident numbers.

HOW WE READ YOUR SUBMISSION

`Correctness · 30%`	Do the numbers hold up, and did you say what definitions you used to get them?
`Skepticism · 25%`	Question 5. What did you find, and did it change any of your other answers?
`Engineering · 25%`	Inside the budget, reproducible, and readable by whoever inherits it.
`Writing · 20%`	Five clear answers, with the uncertainty attached to the right ones.

PRACTICALITIES

- **Send us:** one script or notebook that produces all five answers, plus a one-page write-up with the numbers, your peak memory and your runtime.
- **Use AI freely.** We do, all day. Just tell us what you generated and what you actually checked yourself.
- **If you have time:** what would you change at 100x this size? The real table is 11.4M rows and grows daily.